Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/04/2024
PI Name:	Santa Maria, Sergio Ph.D.		
Project Title:	Acquisition of Beneficial Mutations Through Adaptive Evolution Under Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sergio.r.santamaria@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1411
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Research Branch		
PI Address 2:	MS 288-2		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	06/01/2021	**End Date:**	12/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per F. Hernandez/ARC (Ed., 10/27/23) NOTE: End date changed to 12/31/2023 per F. Hernandez/ARC (Ed., 6/2/23) NOTE: End date changed to 02/28/2023 per F. Hernandez/ARC (Ed., 4/25/22)		
Key Personnel Changes/Previous PI:	Co-Investigator (Co-I) Lauren Liddell is no longer part of the project due to maternity leave starting mid-2022. An undergraduate student intern was recruited in the Fall 2022, Spring 2023, and Fall 2023 to support the Principal Investigator (PI); the student was offered a permanent position starting Jan 2024.		
COI Name (Institution):	Sherlock, Gavin Ph.D. (Stanford University)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Adaptive evolution plays a major role in the appearance of new phenotypes in nature and is of key importance in biomedical research, including the onset of carcinogenesis and in the emergence of drug-resistant microorganisms. The harsh environment of space provides the ideal environment for the acquisition of adaptive mutations, particularly in microbial organisms. Experimental evolution studies in the laboratory will allow us to better understand how specific mutations arise under spaceflight conditions. Our primary goal is to investigate how a eukaryotic organism responds to simulated microgravity after multiple generations. S. cerevisiae or budding yeast is a single-cell model eukaryote that has been widely used in biomedical research, primarily due to its well-characterized biology, easy manipulation, short division time, and, most importantly, its similarities with human cell functions. We hypothesize that beneficial mutations acquired through multiple generations under simulated microgravity will render cells with increased growth and/or metabolic rates, and phenotypes different to those observed in 1g controls. Aim1 involves the identification of adaptive beneficial mutations via experimental evolution and next-generation sequencing. Barcoded yeast libraries will be grown using rotating wall vessels (RWVs) for multiple generations. DNA barcodes are unique 20-nucleotide sequences that are used to “tag” or “barcode” individual strains and allow us to follow the appearance of these adaptive mutations and to identify which mutations are actually beneficial to survive under simulated microgravity. The frequency of the resulting yeast lineages and the appearance of adaptive mutations can be then followed by genomic DNA extraction followed by amplification and sequencing. Aim 2 involves the generation of mutational spectra from the surviving yeast at different time points during multigenerational growth. We propose to perform mutation analyses via whole genome sequencing to generate mutation spectra after long-term exposure to simulated microgravity. The resulting acquired mutations will be then compared to those of the non-exposed 1g controls and to previous generation spectra. All RWV growth studies will be performed at Ames Research Center. Barcoded yeast libraries developed by the Sherlock team at Stanford University are already available. DNA amplification and next-generation sequencing will be performed at Stanford. All project data will be archived in the NASA GeneLab repository. Studies like the one proposed here will allow us to identify beneficial mutations responsible for an adaptive response to space-like conditions. The proposed work will address questions on how spaceflight affects microbial growth and population dynamics (Space Biology Microbiology objectives A-C), and how it alters DNA function and intracellular processes (Cell and Molecular Biology objectives B and E). We will try to identify what processes are influenced by the spaceflight environment, and whether long-term exposure has biological effects across multiple generations (Guiding Questions MB-1, MB-2, MB-5, MB-6, CMB- 1, CMB-7, and Decadal Survey recommendations P1, P2, CC9, and AH16, https://).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of exploring the effects of the space environment on microbes (and their adaptation) has important implications. For example, abnormal cell growth and physiology – especially in pathogens – could be detrimental to astronaut health, especially considering NASA’s upcoming long-term missions to outer space. On the other hand, altered microbial growth in space could be advantageous for the production of high-value products, including medicines, vitamins, and food.
Task Progress:	Specific Aim 1: To investigate the acquisition of adaptive beneficial mutations via experimental evolution and high-throughput sequencing. After optimizing the yeast growth conditions, we performed the 21-day evolution experiment under simulated microgravity using rotating wall vessels (RWVs). The RWV experiments were concluded at the end of October 2023 without issues. Yeast samples were collected every 24 hours (approximately every 8 generations) and stored at either -20°C (sorbitol solutions) or -80°C (glycerol stocks). The samples were then transferred to the Stanford University Co-Investigator in November, where they are currently undergoing genomic DNA extraction, amplification, and sequencing. Sequencing of the yeast samples at different generations or time points will allow us to follow DNA barcodes contained within the yeast library. These unique barcodes are used to tag individual yeast lineages and to follow the appearance of adaptive mutations throughout multiple generations. The goal is to identify mutations that allow for better growth under simulated microgravity (i.e., beneficial mutations). Specific Aim 2: To explore the mutational spectrum from the surviving yeast at different time points during multigenerational growth. Once sequencing of samples from a few time points is completed, we will perform mutation analyses via whole-genome sequencing to generate beneficial mutational spectra. The spectra will be compared to those from the non-exposed controls and from previous generations. Thus, we hope to learn which mutations promote growth under these conditions and how early they appear in the evolution experiments.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Other Journals or Periodicals	Rahmanian S, Slaba T, Straume T, Bhattacharya S, Santa Maria SR. "Meet BioSentinel: the first biological experiment in deep space." Frontiers for Young Minds. 2024 March 6;12:1227860. https://kids.frontiersin.org/articles/10.3389/frym.2024.1227860 , Mar-2024
Articles in Peer-reviewed Journals	Lapointe MR, Laframboise T, Pirkkanen J, Tai TC, Lees SJ, Santa Maria SR, Tharmalingam S, Boreham DR, Thome C. "Protracted exposure to a sub-background radiation environment negatively impacts the anhydrobiotic recovery of desiccated yeast sentinels." Health Phys. 2024 Jun 1;126(6):397-404. http://dx.doi.org/10.1097/HP.0000000000001804 ; PMID: 38568172 , Jun-2024